UNIT 1 · RAPID REVISION

Formula and Fact Sheet

One-page formula & fact sheet

Item	Value / formula
Farm power availability (2021–22 → target)	~3.04 → 4.0 kW/ha (2030)
Overall mechanization level, India	~45–47%
Draft	Pull × cos θ
Drawbar power	Draft (kN) × speed (m/s) = kW
Unit draft	Draft ÷ furrow area (W×d per bottom × n)
Disc plough disc/tilt angle	42–45° / 15–25°
MB plough share suction	vertical (penetration), horizontal (width)
Knife section pitch (mower)	76.2 mm; lead of cutter bar ≈ 2% of length
Cylinder peripheral speed — wheat	25–30 m/s (5000–6000 fpm)
Reel index (combine)	1.25–1.5
Nozzle discharge vs pressure	$q \propto \sqrt{P}$
ULV rate	< 5 L/ha
Droplet uniformity	VMD/NMD → 1
Rice transplanter row spacing	238 mm (fixed, mat type)
Seed & fertilizer band placement	5 cm below × 5 cm beside seed
Happy Seeder purpose	wheat sowing into paddy residue (zero till + straw handling)
Hydraulic agitation nozzles	aimed 30° above horizontal (Kepner §21.16)
Mechanical agitator paddles	100–200 rpm; tip speed <500 fpm (foaming)
FMTTI HQ / oldest	Budni, MP (1955)
Thresher safety / chaff cutter safety	IS 9020 / IS 7898; Dangerous Machines Act 1983
SMAM launch	2014–15
Tractor sales India 2025	~1.0 million units (world's largest)

Further reading

- NPTEL video courses (IIT Kharagpur): "Farm Machinery" and "Tractors and Agricultural Machinery".
- ICAR e-courses (e-Krishi Shiksha): Farm Machinery & Equipment I and II.
- Mechanization data and testing standards: agriwelfare.gov.in (Mechanization & Technology Division); PIB releases on SMAM.
- Kepner, Bainer & Barger — Principles of Farm Machinery, chapters 7–9, 11–15, 17, 19–21.